

Cloud Infrastructure Lab Practicals

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Cloud Lab</title>
</head>
<body>
  <h1>Hello from Container</h1>
  <p>Docker Kubernetes Lab</p>
</body>
</html>
```

Dockerfile

```
FROM nginx
COPY index.html /usr/share/nginx/html/
```

Build Docker Image

```
docker build -t myapp .
```

Create Container

```
docker run -d -p 8080:80 myapp
```

Create 3 Containers

```
docker run -d -p 8081:80 --name c1 myapp
docker run -d -p 8082:80 --name c2 myapp
docker run -d -p 8083:80 --name c3 myapp
```

Port Mapping

```
docker run -d -p 8085:80 myapp
```

Docker Commands

```
docker ps
docker ps -a
docker images
docker build -t myapp .
```

```
docker run -d myapp
docker run -d -p 8080:80 myapp
docker stop container_id
docker rm container_id
docker pull nginx
```

Kubernetes Pod

```
kubectl run mypod --image=myapp
```

Kubernetes Deployment 3 replicas

```
kubectl create deployment mydeploy --image=myapp --replicas=3
```

Kubernetes Commands

```
kubectl get pods
kubectl get nodes
kubectl get deployments
kubectl delete pod name
kubectl delete deployment name
kubectl describe pod name
kubectl apply -f pod.yaml
```

pod.yaml

```
apiVersion: v1
kind: Pod
metadata:
  name: mypod
spec:
  containers:
  - name: mycontainer
    image: myapp
    ports:
    - containerPort: 80
```

deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: mydeploy
spec:
  replicas: 3
  selector:
    matchLabels:
      app: myapp
```

```
template:
  metadata:
    labels:
      app: myapp
  spec:
    containers:
      - name: mycontainer
        image: myapp
        ports:
          - containerPort: 80
```